



NANJING SHIHENG ELECTRONICS CO., LTD.

MF72 Series Inrush Current Limiter NTC Thermistors



MF72 Series Inrush Current Limiter NTC Thermistors

Precision · Stability · Reliability

Designed to suppress high inrush current in power circuits, the MF72 Series ensures reliable protection, excellent stability, and extended system lifespan.

Featuring robust silicon resin coating and wide resistance range, it provides precise current control across various power applications.

Issue Date: January 2026

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Specifications and data are subject to change without prior notice.

Please contact us for the latest specifications.

Features

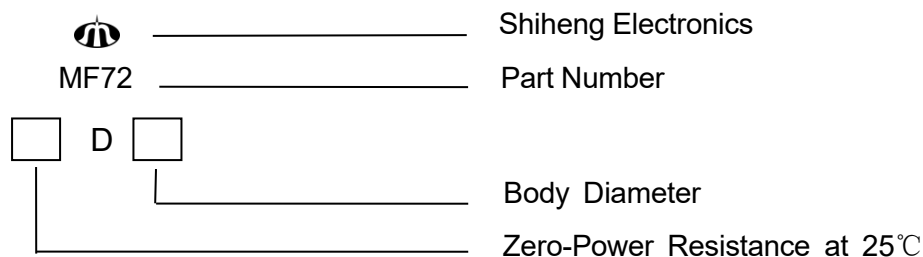
- Compliant with RoHS, Halogen-Free, and REACH requirements
- Body size: $\varnothing 5$ mm to $\varnothing 25$ mm
- Radial-leaded, silicon resin coated construction
- Operating temperature range: -40°C to $+200^{\circ}\text{C}$
- Safety certifications: UL / cUL / TÜV / CQC

Applications

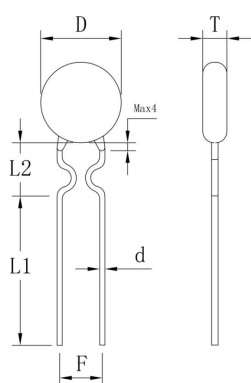
- Server and communication power supplies
- UPS and backup power systems
- High-power LED lighting circuits
- Industrial and commercial lighting drivers
- Inverters
- High-power AC / DC converter systems
- Industrial control power supplies
- Power tools and portable charging equipment
- Professional high-power audio amplifiers



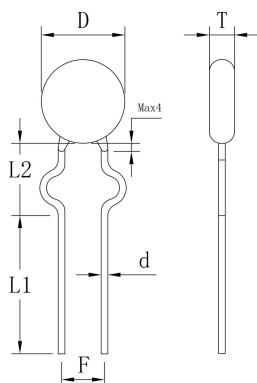
Part Number Construction



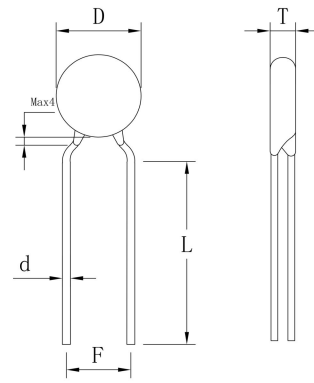
Structure and Dimension



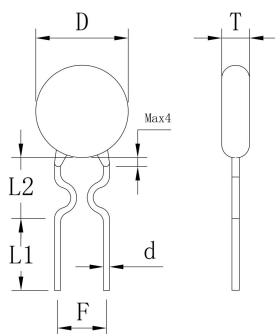
Inward-Formed Long Leads



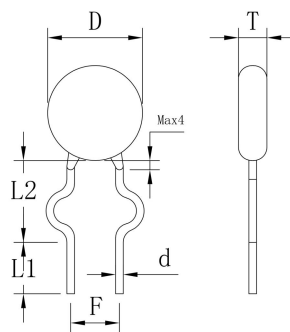
Outward-Formed Long Leads



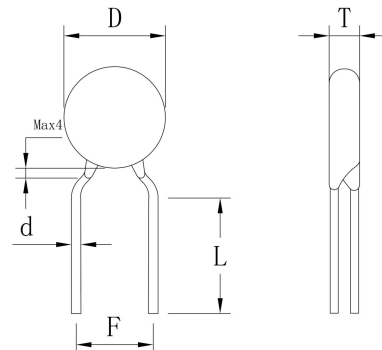
Shoulder-Formed Long Leads



Inward-Formed Short Leads



Outward-Formed Short Leads



Shoulder-Formed Short Leads

Mechanical Specifications (Unit: mm)

Part No.	D _{max}	T _{max}	d ± 0.05	F ± 0.5	L _{min}	Formed Leads	
						L1 _{min}	L2 ± 2
MF72 □D5	6.0	5.0	0.45/0.6	2.5/5.0	25	20/3.5	5
MF72 □D7	8.0	5.0	0.6	5	25	20/3.5	5
MF72 □D9	10.0	5.5	0.8	7.5/5	25	20/3.5	5
MF72 □D11	12.0	5.5	0.8	7.5/5	25	20/3.5	5
MF72 □D13	14.0	6.0	0.8	7.5	25	20/3.5	5
MF72 □D15	16.0	6.0	0.8	7.5	25	20/3.5	5
MF72 □D20	21.0	7.0	1.0	10	25	20/4.0	8
MF72 □D25	27.5	8.0	1.0	10	25	20/4.0	8

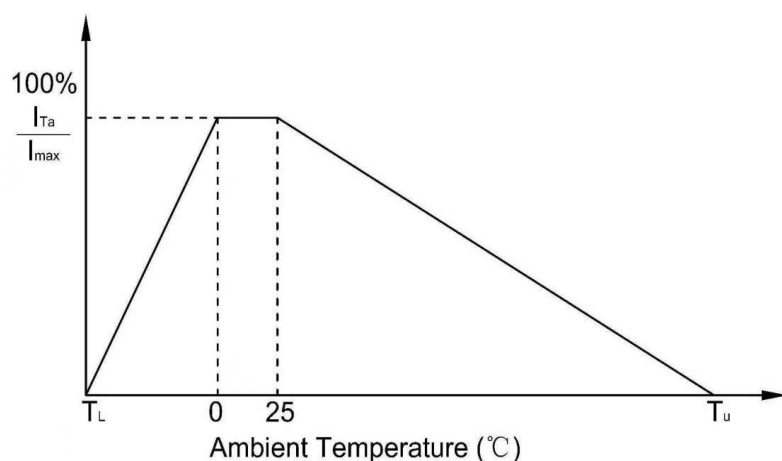
Electrical Specifications

Part No.	R ₂₅ ± 20%	B Value (25/50℃) ± 10%	Maximum Steady-State Current	Residual Resistance (Approx)	Max. Capacitance @ 240 Vac	Max. Capacitance @ 120 Vac	P _{max}	Operating Temperature Range	Certifications		
	Ω	K	A	Ω	μF	μF	W	T _L ~ T _U (℃)	UL / cUL	TÜV	CQC
MF72 5D5	5	2500	1	0.403	100	400	0.5	-40 ~ +150	√	√	√
MF72 10D5	10	2600	0.7	0.771	100	400			√	√	√
MF72 5D7	5	2600	2	0.210	150	600	1		√	√	√
MF72 10D7	10	2800	1	0.618	100	400			√	√	√
MF72 16D7	16	2800	0.7	1.522	68	272			√	√	√
MF72 3D9	3	2600	4	0.092	220	880	2	-40 ~ +170	√	√	√
MF72 5D9	5	2600	3	0.153	220	880			√	√	√
MF72 8D9	8	2800	2	0.336	150	600			√	√	√
MF72 10D9	10	2800	2	0.280	150	600			√	√	√
MF72 3D11	3	2600	5	0.079	470	1880	3		√	√	√
MF72 5D11	5	2800	4	0.109	330	1320			√	√	√
MF72 8D11	8	2800	3	0.185	220	880			√	√	√
MF72 10D11	10	2800	3	0.212	220	880			√	√	√
MF72 16D11	16	3000	2	0.361	220	880			√	√	√
MF72 3D13	3	2600	6	0.072	560	2240	4	-40 ~ +200	√	√	√
MF72 5D13	5	2800	5	0.101	560	2240			√	√	√
MF72 8D13	8	3000	4	0.138	330	1320			√	√	√
MF72 10D13	10	3000	4	0.144	330	1320			√	√	√
MF72 16D13	16	3200	3	0.279	330	1320			√	√	√
MF72 3D15	3	2800	7	0.069	680	2720	5		√	√	√
MF72 5D15	5	3000	6	0.093	680	2720			√	√	√
MF72 8D15	8	3000	5	0.133	470	1880			√	√	√
MF72 10D15	10	3200	5	0.146	470	1880			√	√	√
MF72 16D15	16	3200	4	0.211	470	1880			√	√	√
MF72 3D20	3	2800	8	0.071	820	3280	6		√	√	√
MF72 5D20	5	3000	7	0.097	820	3280			√	√	√
MF72 8D20	8	3000	6	0.131	680	2720			√	√	√
MF72 10D20	10	3200	6	0.146	680	2720			√	√	√
MF72 16D20	16	3200	5	0.199	680	2720			√	√	√
MF72 3D25	3	2800	9	0.069	1000	4000	7.5	-40 ~ +200	√	√	√
MF72 5D25	5	3000	8	0.092	1000	4000			√	√	√
MF72 8D25	8	3200	7	0.123	820	3280			√	√	√
MF72 10D25	10	3200	7	0.133	820	3280			√	√	√
MF72 16D25	16	3200	6	0.167	820	3280			√	√	√

Notes:

1. For products D5 to D11, the lead wire material is tinned CP wire; for D13 to D25, the lead wire material is tinned copper wire.
2. Coating Material:
D5 to D15 – Green silicone resin
D20 / D25 – Black silicone resin
3. Certifications:
UL / cUL Certificate No.: E241319
TÜV Certificate No.: R50245892
CQC Certificate No.: CQC04001010556
4. For special requirements, please contact our sales department.

Current Derating Curve



Example:

Ambient temperature (T_a) = -10°C

Minimum operating temperature (T_L) = -40°C

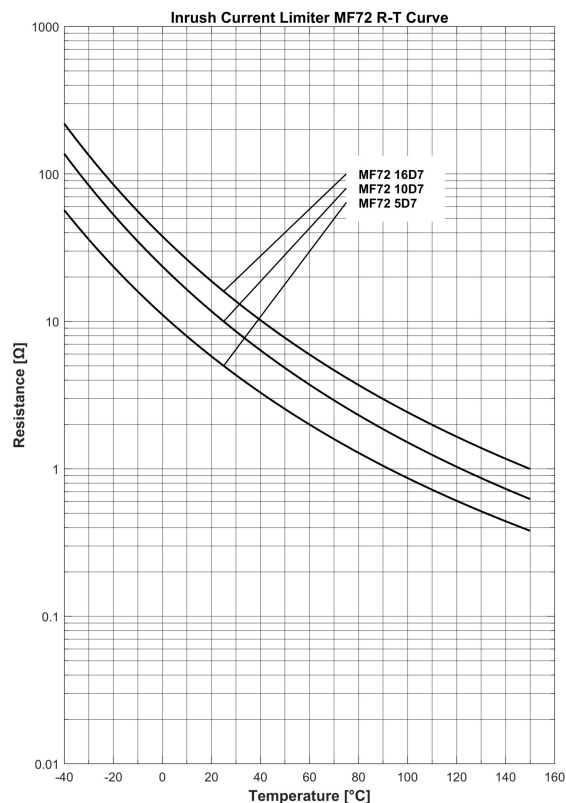
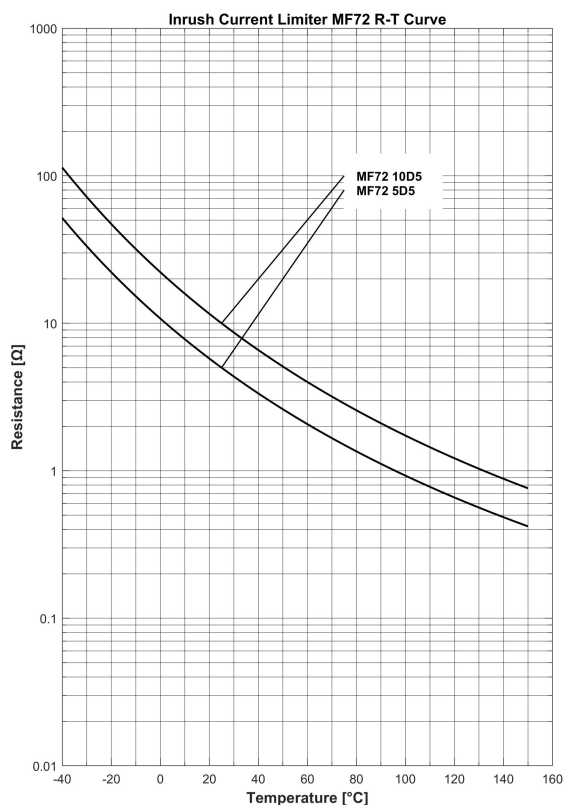
$$I_{Ta} = \frac{T_L - T_a}{T_L - 0} \times I_{max} = 75\% I_{max}$$

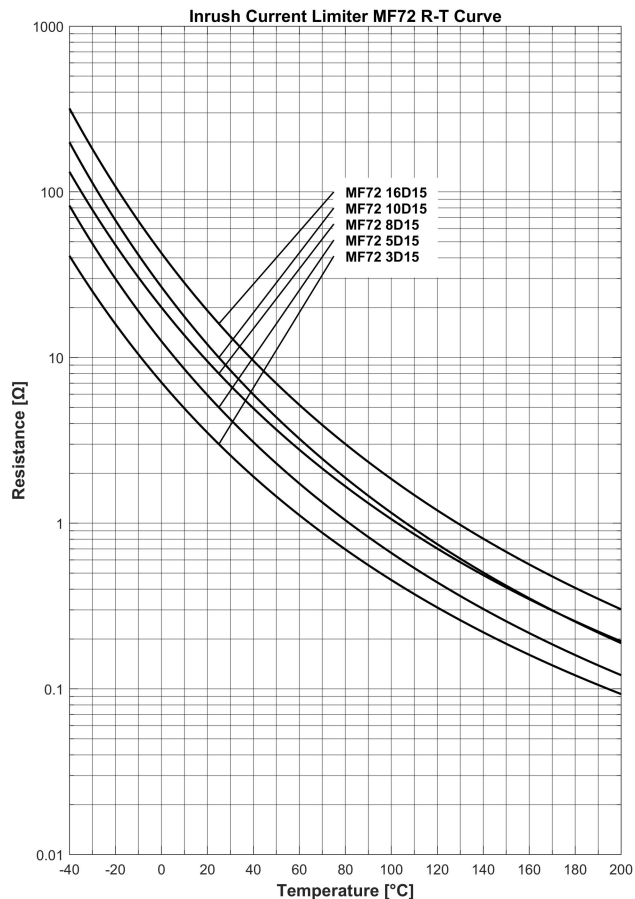
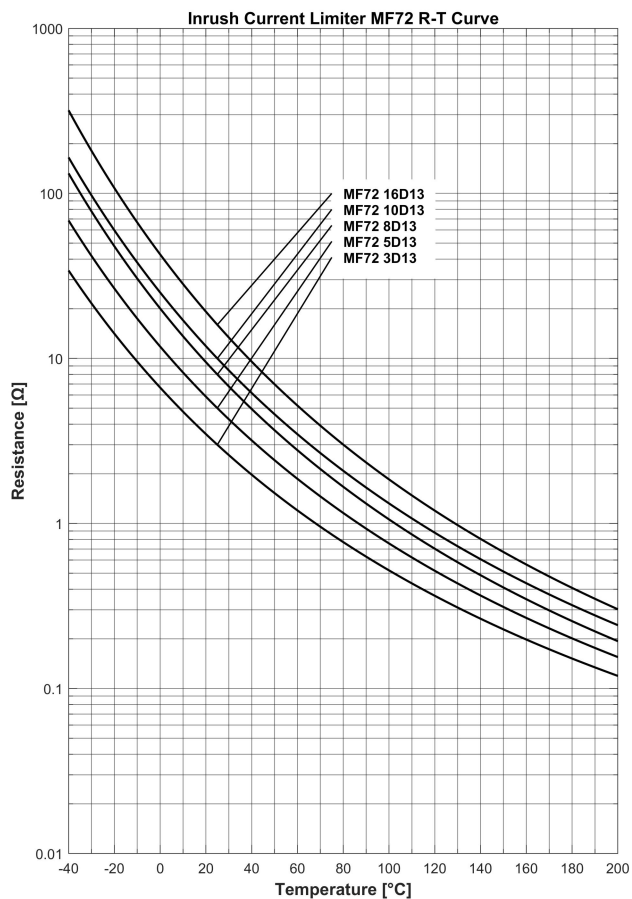
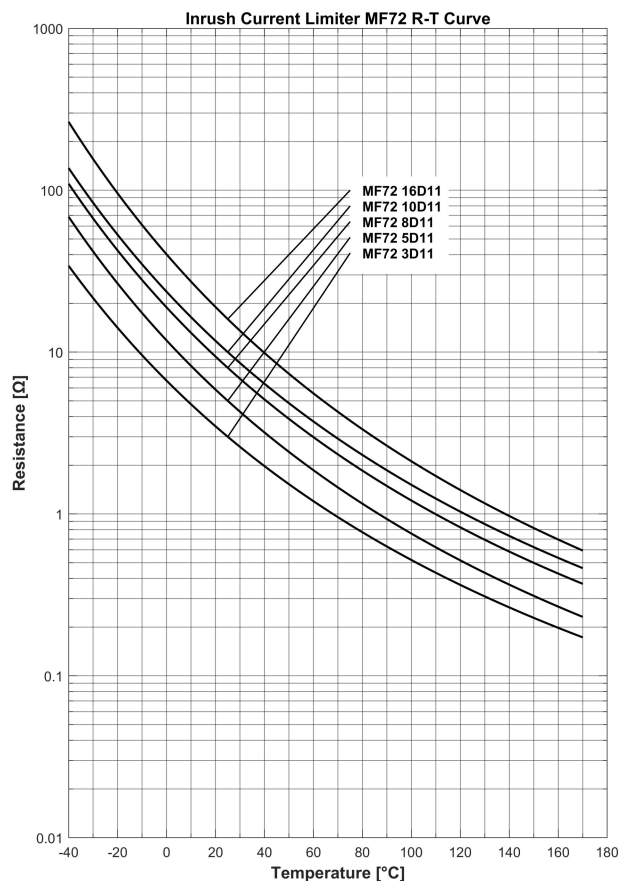
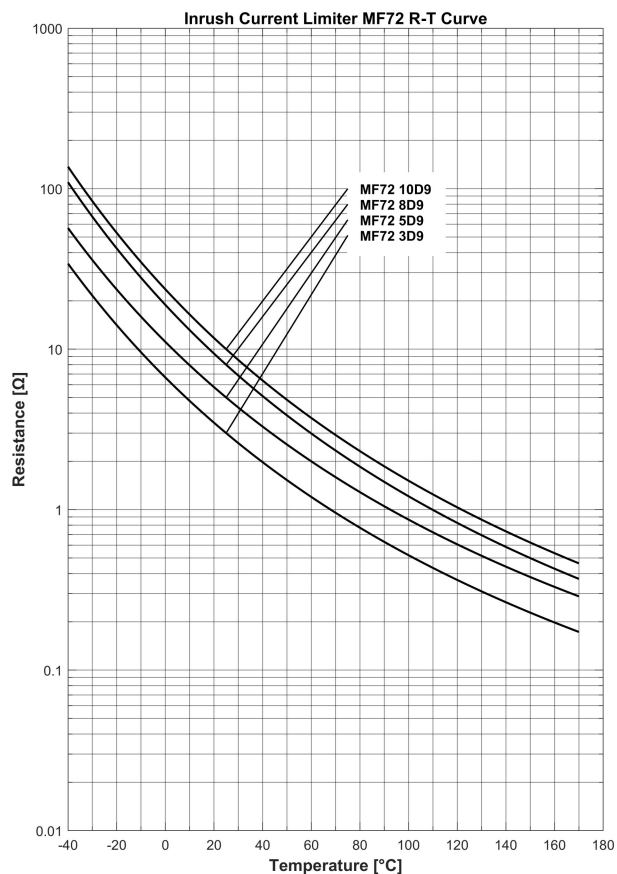
Ambient temperature (T_a) = 55°C

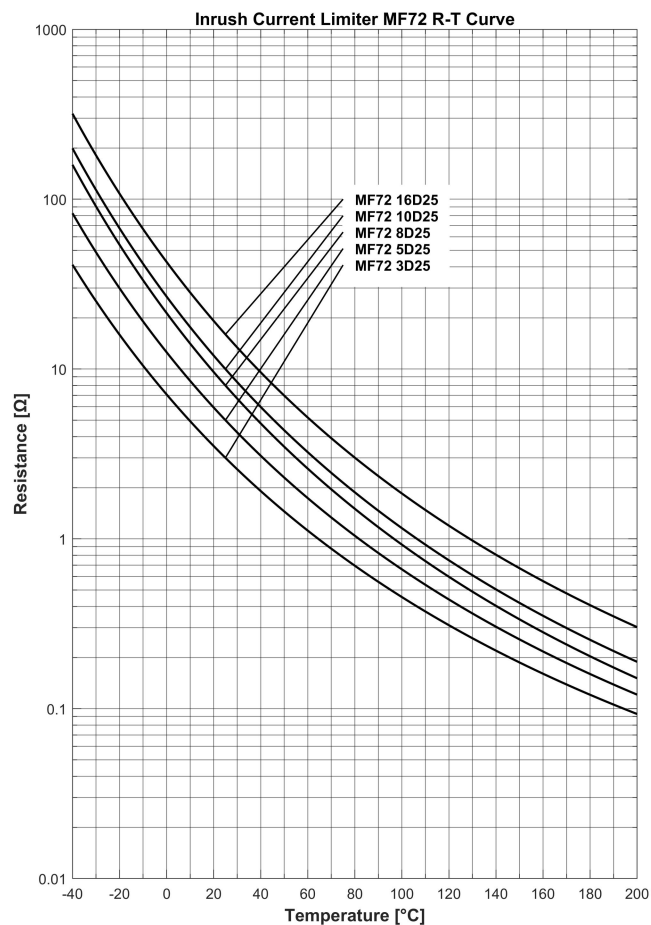
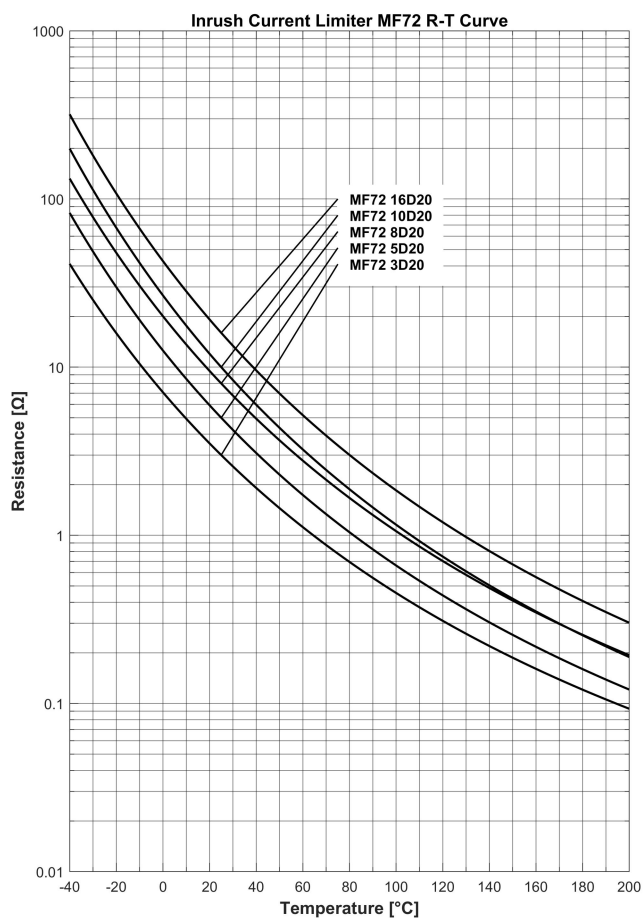
Maximum operating temperature (T_u) = 200°C

$$I_{Ta} = \frac{T_u - T_a}{T_u - 25} \times I_{max} = 83\% I_{max}$$

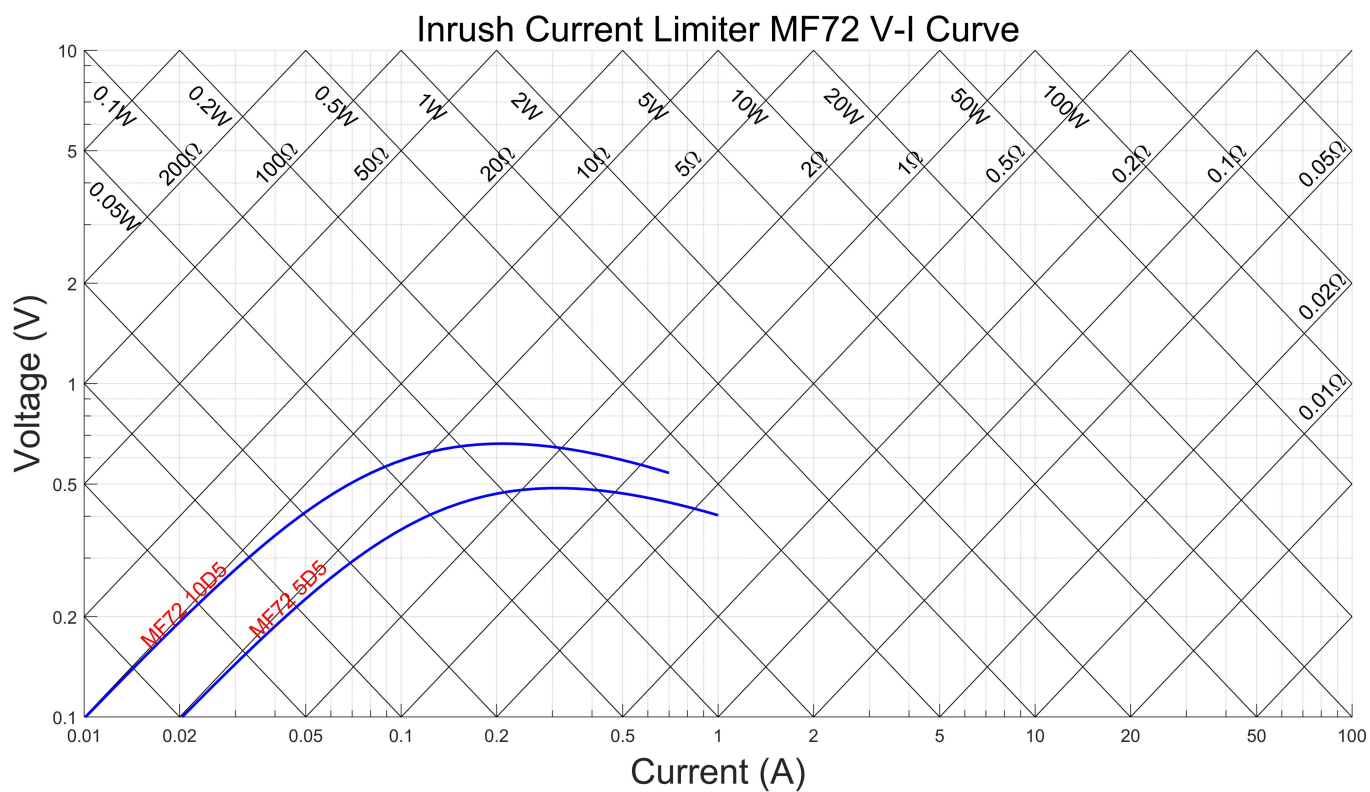
R-T characteristic Curves

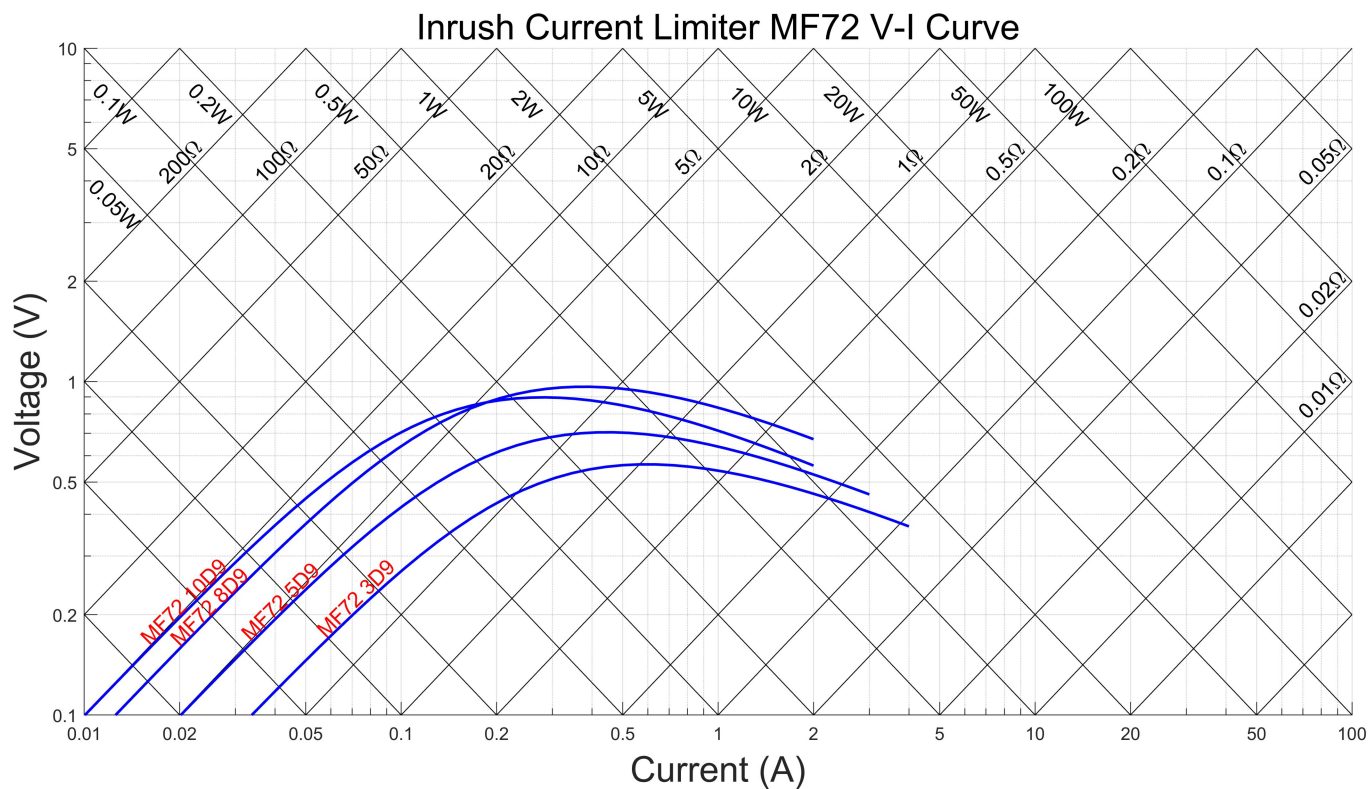
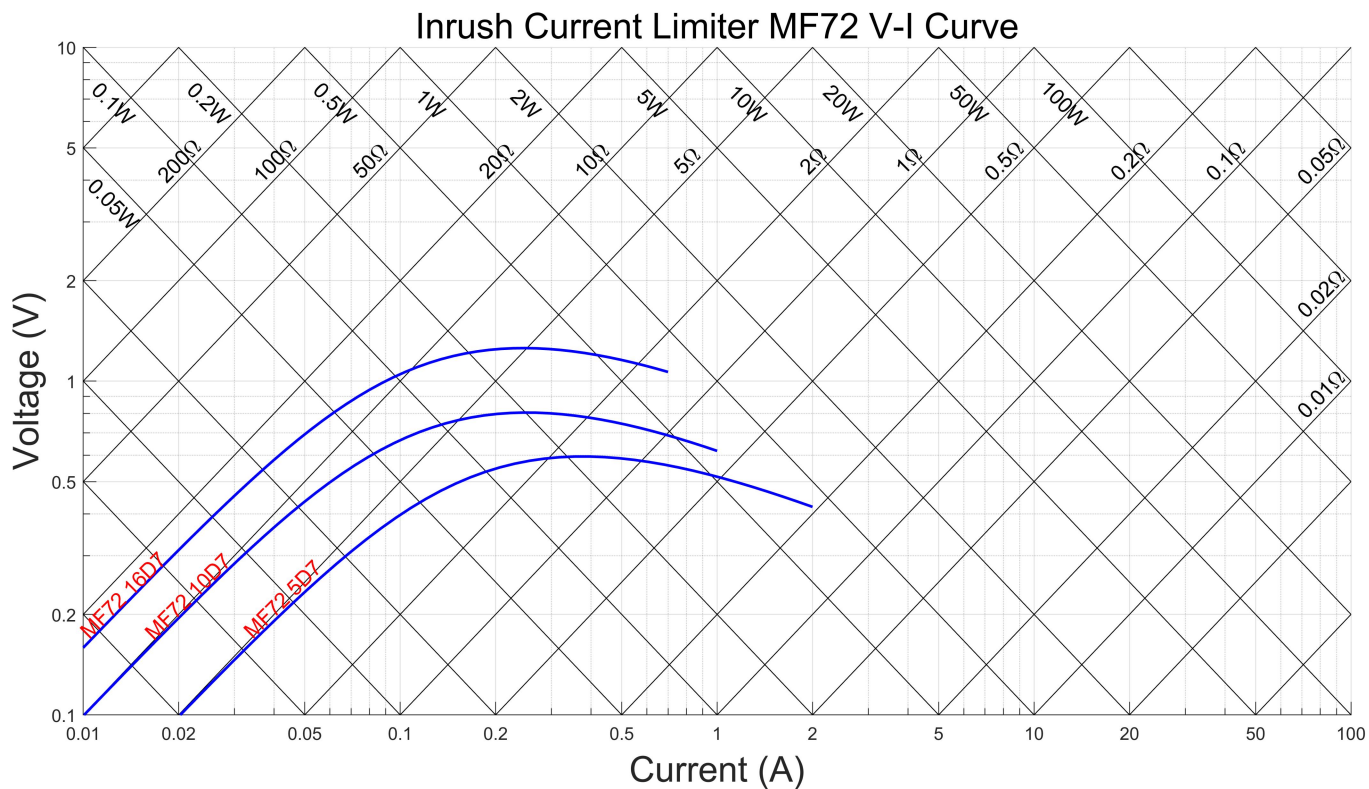


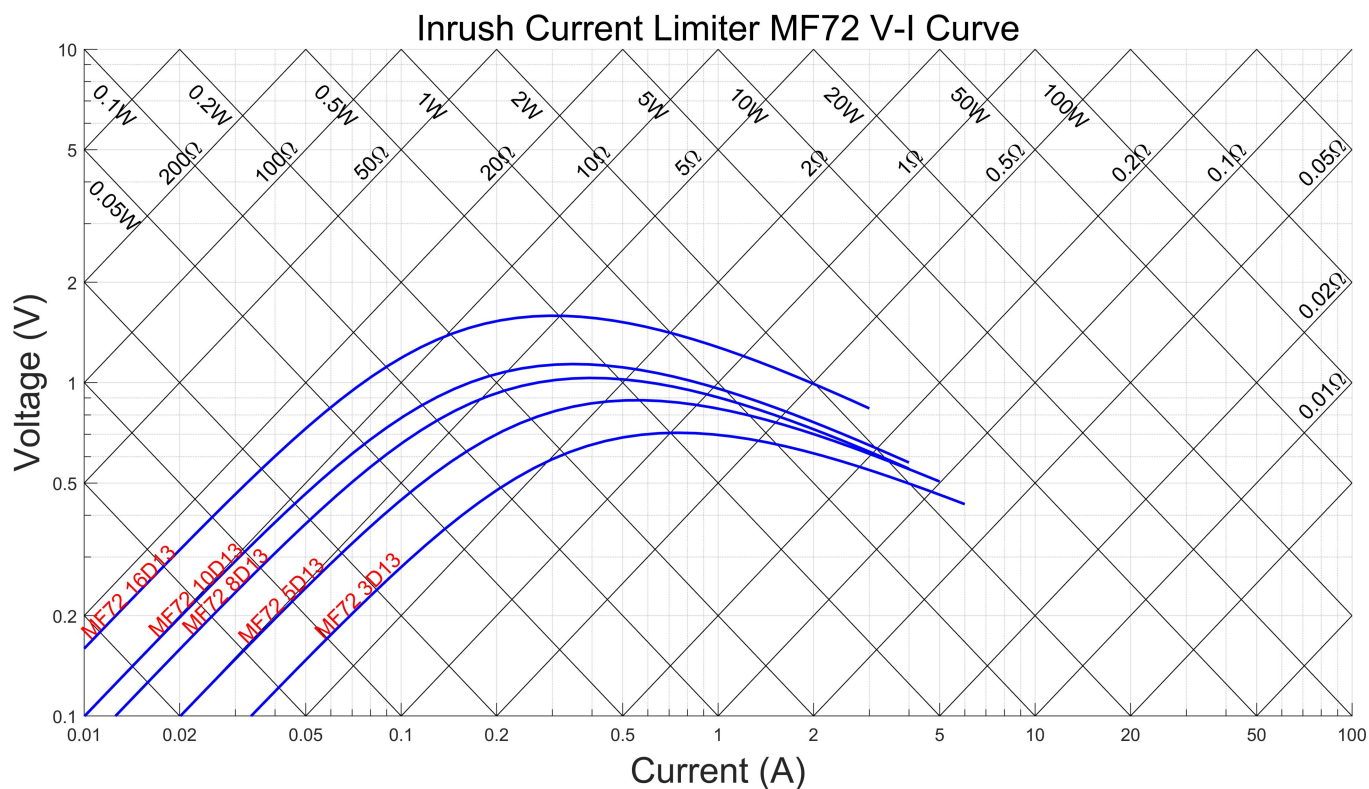
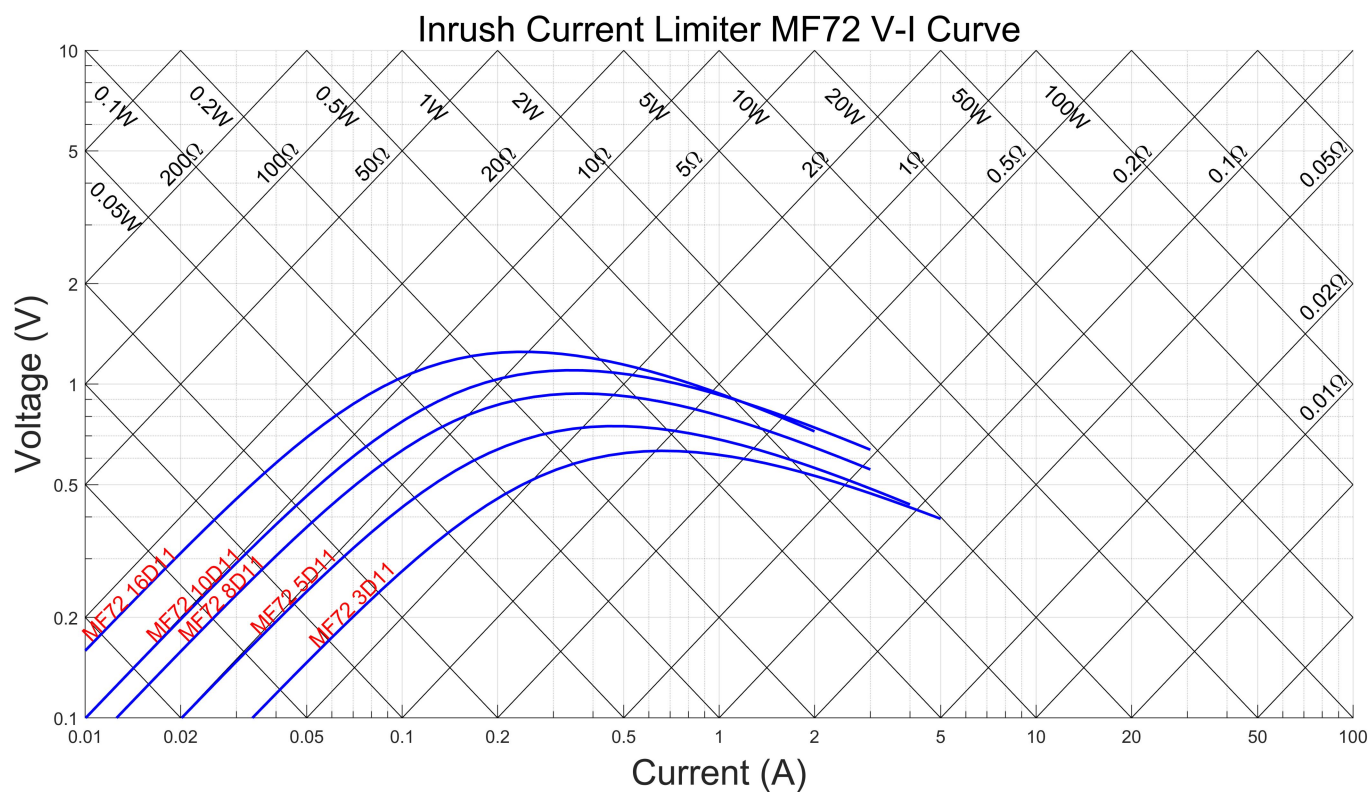


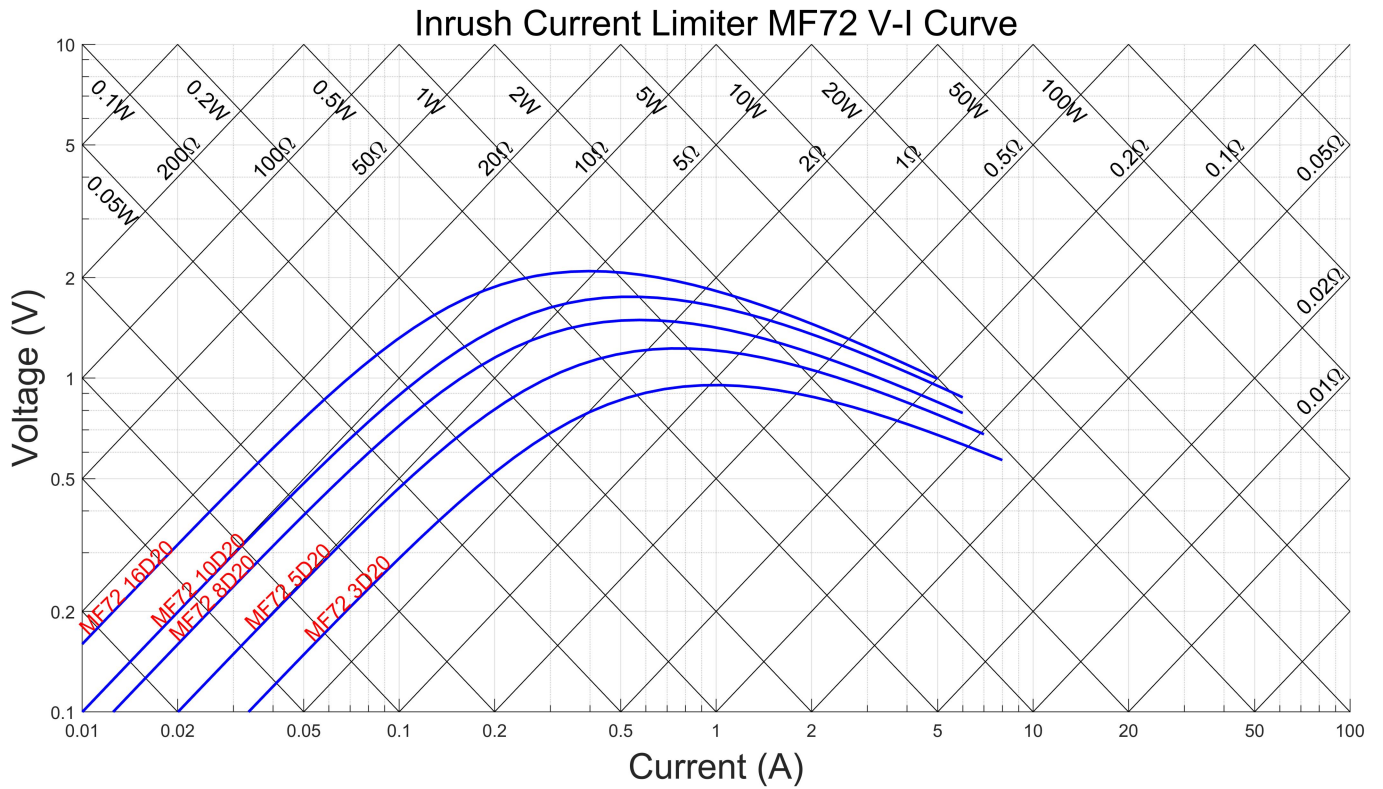
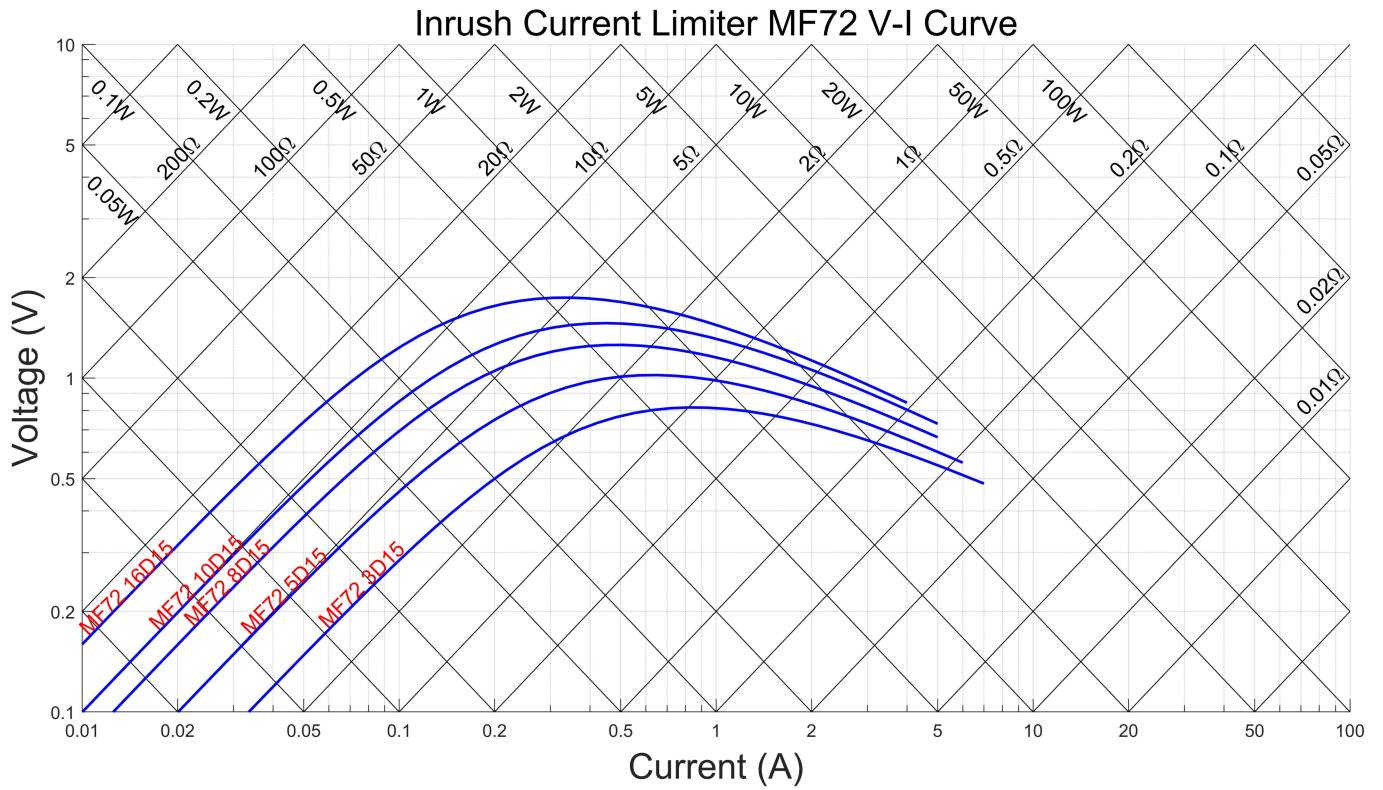


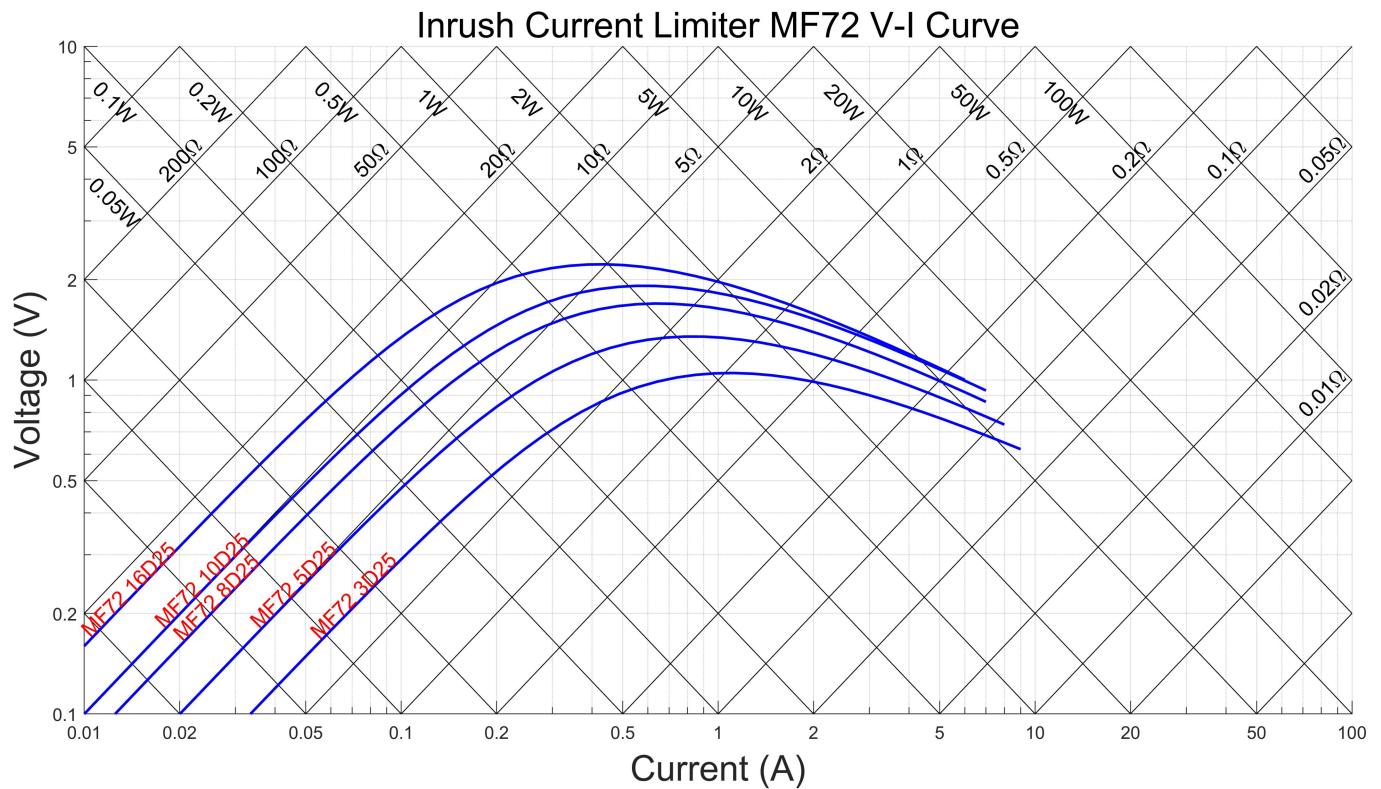
V-I Characteristics Curve



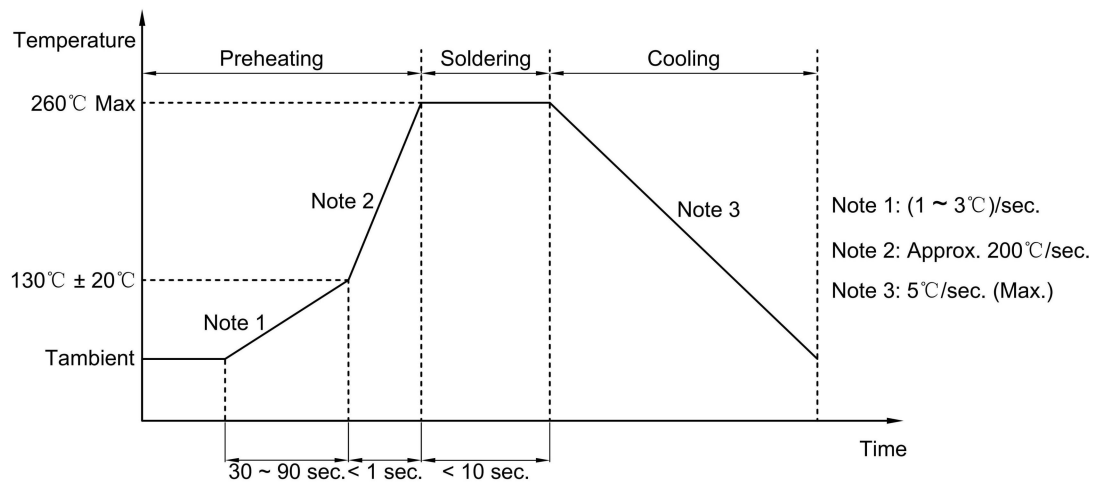








Wave Soldering Profile



Recommended Reworking Conditions With Soldering Iron

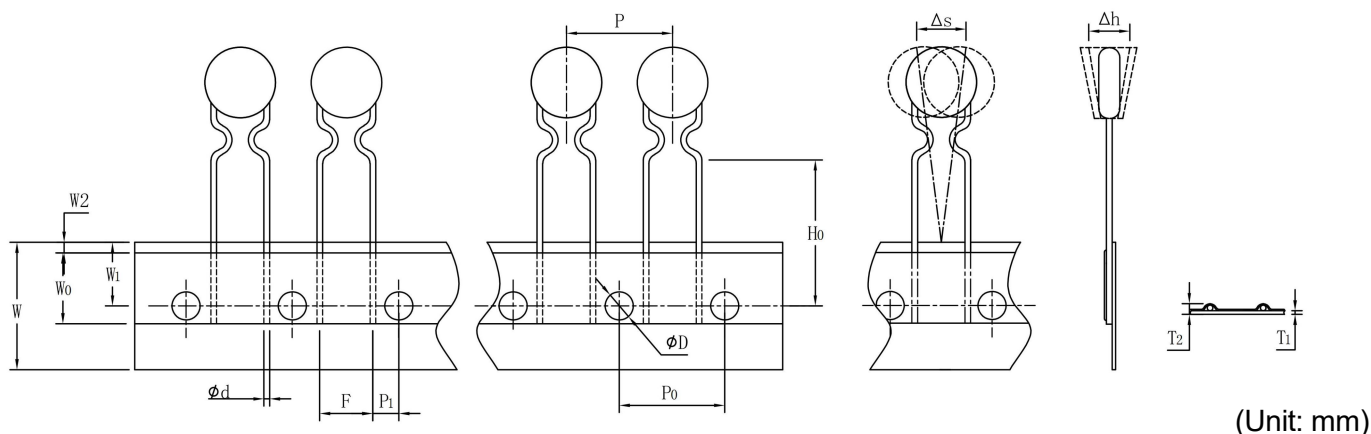
Item	Condition
Temperature of Soldering Iron Tip	360°C (max.)
Soldering Time	3 sec (max.)
Distance from Thermistor	2 mm (min.)

Reliability

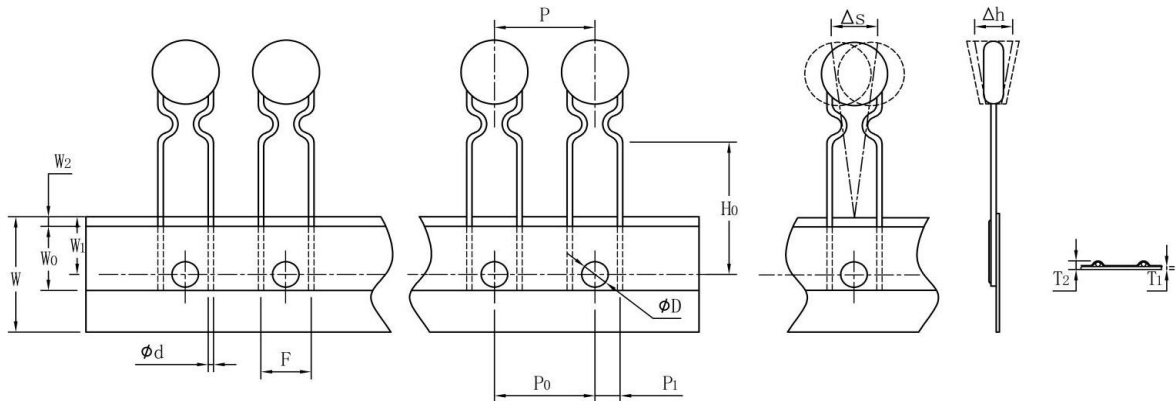
No.	Test Item	Test Standard	Test Conditions / Methods	Acceptance Criteria								
1	Terminal strength	IEC60068-2-21	Gradually apply the specified force and keep the unit fixed for 10 ± 1 sec. <table><thead><tr><th>Lead diameter (mm)</th><th>Pull strength (N)</th></tr></thead><tbody><tr><td>$d \leq 0.5$</td><td>5</td></tr><tr><td>$0.5 < d \leq 0.8$</td><td>10</td></tr><tr><td>$0.8 < d \leq 1.25$</td><td>20</td></tr></tbody></table> Duration: 10 ± 1 seconds.	Lead diameter (mm)	Pull strength (N)	$d \leq 0.5$	5	$0.5 < d \leq 0.8$	10	$0.8 < d \leq 1.25$	20	No visible damage. $ \Delta R_{25} / R_{25} \leq 20\%$
Lead diameter (mm)	Pull strength (N)											
$d \leq 0.5$	5											
$0.5 < d \leq 0.8$	10											
$0.8 < d \leq 1.25$	20											
2	Solderability	IEC60068-2-20	Temperature: $245^{\circ}\text{C} \pm 5^{\circ}\text{C}$, for 2 ~ 3 seconds.	Coverage area $\geq 95\%$								
3	Resistance to Soldering Heat	IEC60068-2-20	Temperature of solder bath: $260^{\circ}\text{C} \pm 5^{\circ}\text{C}$, insertion depth: 6 mm from resistor body, duration: 10 ± 1 seconds.	No visible damage. $ \Delta R_{25} / R_{25} \leq 20\%$								
4	Steady humidity and heat	IEC60068-2-78	Temperature: $40^{\circ}\text{C} \pm 2^{\circ}\text{C}$, 93% RH $\pm 2\%$ RH, 1000 hrs.	No visible damage. $ \Delta R_{25} / R_{25} \leq 20\%$								
5	Rapid Temperature Cycling	IEC60068-2-14	The conditions shown below shall be repeated 5 cycles.		No visible damage. $ \Delta R_{25} / R_{25} \leq 20\%$							
			Step	Temperature ($^{\circ}\text{C}$)		Period (minutes)						
			1	$T_L \pm 5$		30 ± 3						
			2	Room temperature		5 ± 3						
			3	$T_U \pm 5$		30 ± 3						
4	Room temperature	5 ± 3										
6	High Temperature Storage	IEC60068-2-2	Temperature: $T_U \pm 2^{\circ}\text{C}$, Duration: 1000 hrs.		No visible damage. $ \Delta R_{25} / R_{25} \leq 20\%$							
7	Maximum Steady-State Current	IEC60539-1	Apply maximum steady-state current to the thermistor at ambient temperature for 1000 ± 24 hours.		No visible damage. $ \Delta R_{25} / R_{25} \leq 20\%$							
8	Maximum Permissible Capacitance	IEC60539-1	Apply the maximum permissible capacitance to the thermistor, with a duty cycle consisting of 50 ms ON and an OFF period equivalent to five times the thermal time constant. Repeat this cycle 1000 times.		No visible damage. $ \Delta R_{25} / R_{25} \leq 20\%$							

Packing Method

1. Taped packaging (inward Lead)



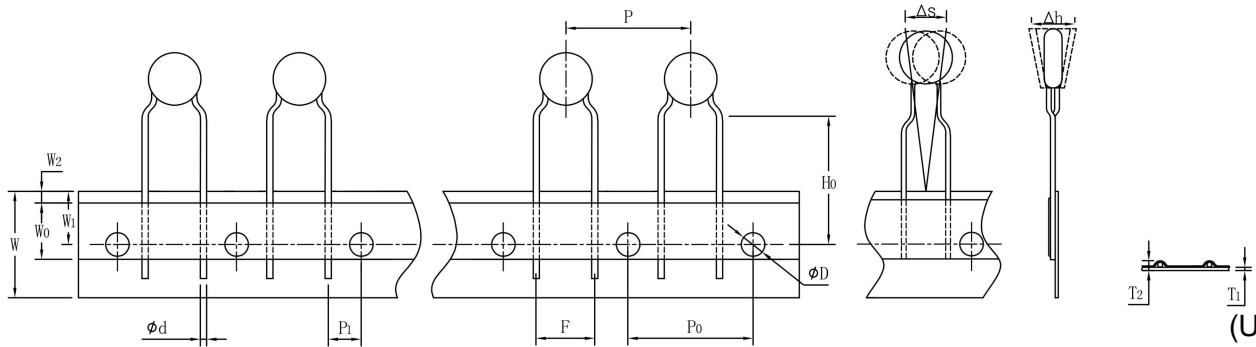
Taping Dimensions	P	F	P ₀	P ₁	W	W ₀	W ₁	W ₂	ØD	T ₁	T ₂	Δh	Δs	Ød	H ₀
	± 1.0	± 0.5	± 0.3	± 0.5	± 0.5	± 2	± 0.5	Max.	± 0.2	± 0.1	Max.	Max.	Max.	± 0.05	+1 / -0
P0: 12.7	12.7	5.0	12.7	3.85	18	10	9	3	4	0.5	1.5	1	1.3	Refer to product	16.0
P0: 15.0	15.0	7.5	15.0	3.75	18	10	9	3	4	0.5	1.5	1	1.3	Refer to product	16.0



(Unit: mm)

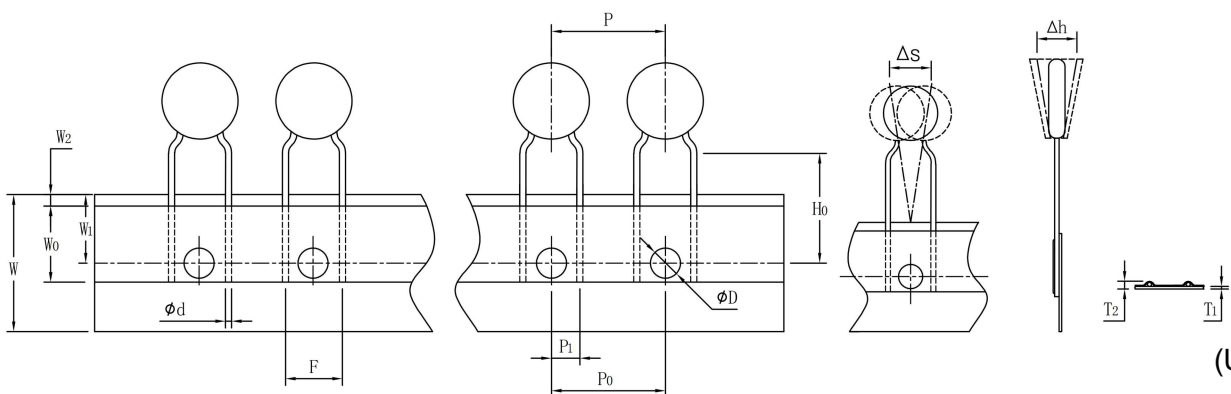
Taping Dimensions	P	F	P ₀	P ₁	W	W ₀	W ₁	W ₂	ØD	T ₁	T ₂	Δh	Δs	Ød	H ₀
	± 1.0	± 0.5	± 0.3	± 0.5	± 0.5	± 2	± 0.5	Max.	± 0.2	± 0.1	Max.	Max.	Max.	± 0.05	+1 / -0
P0: 12.7	12.7	7.5	12.7	3.75	18	10	9	3	4	0.5	1.5	1	1.3	Refer to product	16.0

2.Taped Packaging (Shoulder Leads)



(Unit: mm)

Taping Dimensions	P	F	P ₀	P ₁	W	W ₀	W ₁	W ₂	ØD	T ₁	T ₂	Δh	Δs	Ød	H ₀
	± 1.0	± 0.5	± 0.3	± 0.5	± 0.5	± 2	± 0.5	Max.	± 0.2	± 0.1	Max.	Max.	Max.	± 0.05	+1 / -0
P0: 12.7	12.7	5.0	12.7	3.85	18	10	9	3	4	0.5	1.5	1	1.3	Refer to product	20.0
P0: 15.0	15.0	7.5	15.0	3.75	18	10	9	3	4	0.5	1.5	1	1.3	Refer to product	20.0



(Unit: mm)

Taping Dimensions	P	F	P ₀	P ₁	W	W ₀	W ₁	W ₂	ØD	T ₁	T ₂	Δh	Δs	Ød	H ₀
	± 1.0	± 0.5	± 0.3	± 0.5	± 0.5	± 2	± 0.5	Max.	± 0.2	± 0.1	Max.	Max.	Max.	± 0.05	+1 / -0
P0: 12.7	12.7	7.5	12.7	3.75	18	10	9	3	4	0.5	1.5	1	1.3	Refer to product	20.0

Packaging and Quantity

- Bulk (sealed bag)

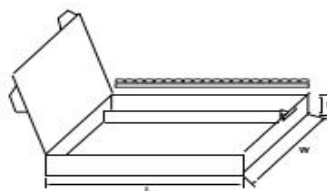
Body Size	Long Lead Quantity (pcs / bag)	Short Lead Quantity (pcs / bag)
Ø5	1000	1000
Ø7	500	1000
Ø9	500	1000
Ø11	500	500
Ø13	300	400
Ø15	200	300

- Box Packaging (Plastic Tray + EPE Foam)

Body Size	Long Lead Quantity (pcs / bag)	Short Lead Quantity (pcs / bag)
Ø20	290	290
Ø25	200	200

- Box Packaging (Taped)

Body Size	Quantity (pcs / box)
Ø5	1500
Ø7	1000
Ø9	1000
Ø11	1000
Ø13	500
Ø15	500



L	W	H
330 mm	240 mm	52 mm

Storage Conditions

- Storage temperature: -10°C to $+40^{\circ}\text{C}$; relative humidity: $\leq 75\%$ RH.
- Avoid storage in environments containing corrosive substances or gases, or exposure to direct sunlight and radiation sources.
- After the packaging is opened, it must be resealed. Shelf life: 1 year.